



University Institute of Engineering

Department of Computer Science & Engineering

Experiment: 2.1

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Subject Name: Probability and Statistics

Subject Code: 25SMH-241

1. Aim of the practical:

To fit a **straight line** to the given set of data points using the **method of least squares** and determine the best-fit equation of the form:

$$y = a + bx$$

where a and b are calculated using the least squares method.

2. Tool/Software Used:

Google Collab

3. Description of the Problem:

To determine the **best-fit straight line** for given data points using the **method of least squares** and obtain the equation $y = a + bx$.

4. Logic:

1. Calculate the number of data points n and the sums $\sum x$, $\sum y$, $\sum x^2$, and $\sum xy$.
2. Calculate the slope b using:
$$b = \frac{n \sum xy - \sum x \sum y}{n \sum x^2 - (\sum x)^2}$$



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3. Calculate the intercept a using:
$$a = \frac{\sum y - b \sum x}{n}$$
4. Obtain the best-fit straight-line equation:
$$Y = a + bX$$
5. Calculate the fitted Y values and the errors between observed and fitted values.
6. Calculate the **sum of squared errors (SSE)** and plot the observed data points along with the best-fit line.

5. Program Code :

```
import numpy as np
import matplotlib.pyplot as plt

# Given data
x = np.array([1, 2, 3, 4, 5])
y = np.array([2, 4, 5, 8, 10])

# Number of data points
n = len(x)

# Calculate required sums
Sx = np.sum(x)
Sy = np.sum(y)
Sx2 = np.sum(x**2)
Sxy = np.sum(x * y)

# Calculate slope 'b'
b = (n * Sxy - Sx * Sy) / (n * Sx2 - Sx**2)

# Calculate intercept 'a'
a = (Sy - b * Sx) / n

# Calculate fitted Y values
Y = a + b * x

# Calculate error
error = y - Y
```



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```
# Sum of squared errors
SSE = np.sum(error**2)

# Display results
print("Value of a =", a)
print("Value of b =", b)

print("\nBest-fit straight line:")
print("Y =", a, "+", b, "X")

print("\nSum of squared errors =", SSE)

# Create values for fitted line
x_line = np.linspace(min(x), max(x), 100)
y_line = a + b * x_line

# Plot observed data points
plt.scatter(x, y, label="Observed Data")

# Plot best-fit line
plt.plot(x_line, y_line, "--", label="Best Fit Line")

plt.xlabel("x")
plt.ylabel("y")
plt.title("Straight Line Fitting by Least Squares Method")
plt.legend()
plt.grid(True)

# Show graph
plt.show()
```

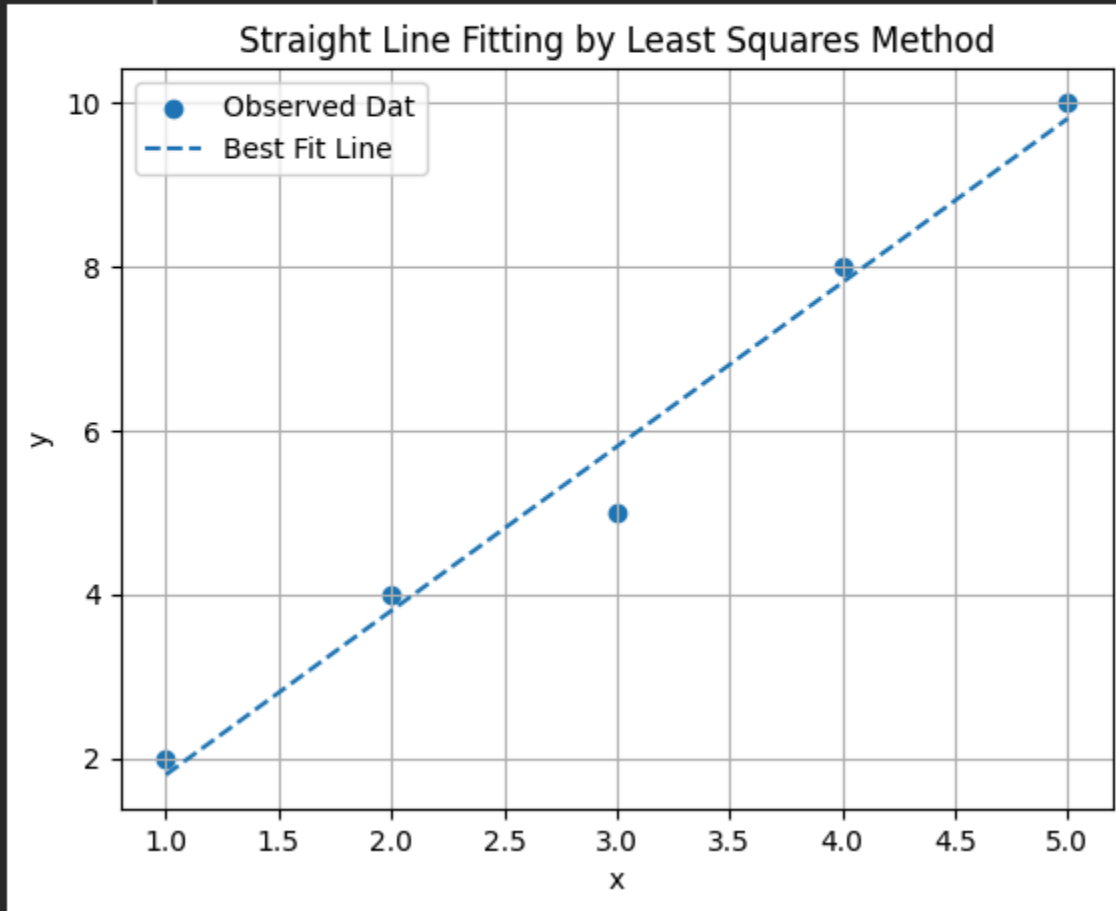
Output:-



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```
... Value of a = -0.2  
Value of b = 2.0  
  
Best-fit straight line:  
Y = -0.2 + 2.0 X  
  
Sum of squared errors = 0.7999999999999995
```



6. Observations, Simulation Screenshots and Discussions:

Observations:

- The given data points are $x=[1,2,3,4,5]$ and $y=[2,4,5,8,10]$.
- The calculated intercept a and slope b give the best-fit straight line.



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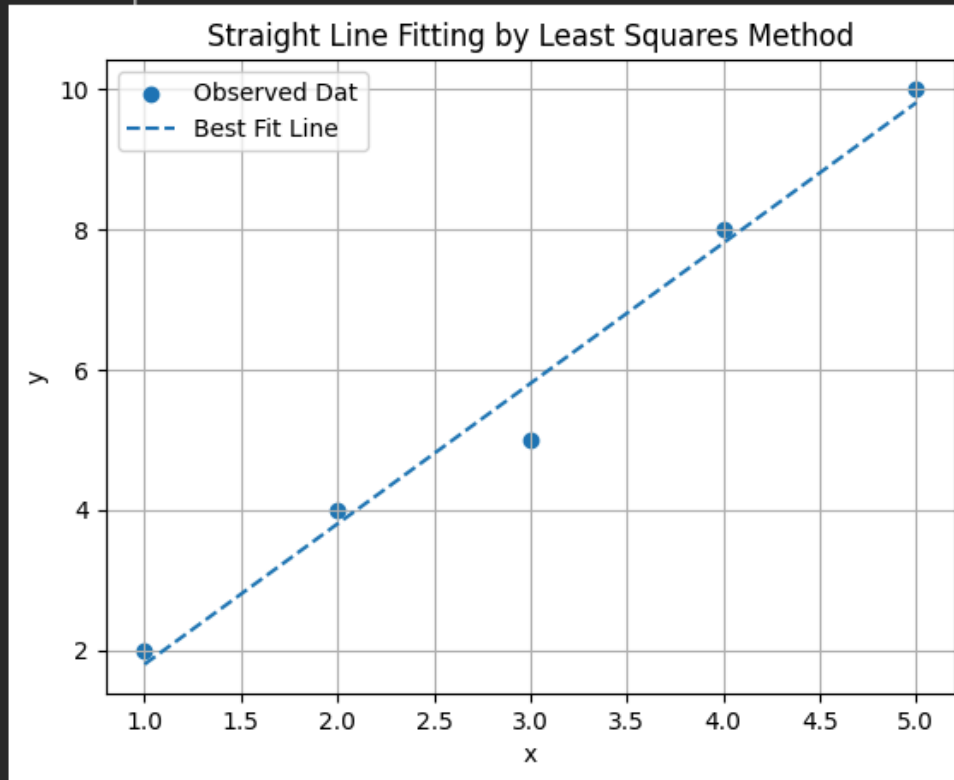
- The observed data points and fitted line are displayed graphically.
- The **Sum of Squared Errors (SSE)** is calculated to measure the fitting error.

Simulation Screenshot:

```
*** Value of a = -0.2  
Value of b = 2.0
```

```
Best-fit straight line:  
Y = -0.2 + 2.0 X
```

```
Sum of squared errors = 0.7999999999999995
```



Discussion:

The least squares method provides a straight line that best represents the given data by minimizing the squared errors between observed and fitted values.



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7 . Result and Summary:

The best-fit straight line was successfully obtained using the **method of least squares**. The slope, intercept, fitted values, and **Sum of Squared Errors (SSE)** were calculated, and the observed data points were plotted along with the fitted straight line.

8. Additional Creative Inputs (If Any):

1. Use different datasets to demonstrate straight-line fitting in real-life situations.
2. Compare the **observed values** with the **fitted values** using a table.
3. Plot the data points and best-fit line to visually analyze the relationship between xx and yy.
4. Calculate and compare the **SSE** for different datasets to determine the quality of the fit.

Learning outcomes (What I have learnt):

1. Understand the **method of least squares**.
2. Calculate the **slope and intercept** of a straight line.
3. Fit a straight line to a given set of data points.
4. Calculate **fitted values and errors**.
5. Represent the best-fit line graphically using **Python**.

Conclusion:

The experiment successfully demonstrated the **curve fitting of a straight line using the method of least squares**. The best-fit line was obtained by calculating the slope and intercept, and the observed data points were compared with the fitted values. The graphical representation helped in understanding the relationship between the variables.